BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Mathematics (Objective) Group - I

Time Allowed: 30 Minutes | **Maximum Marks:** 20 | **Paper Code:** 8197

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. The lines represented by $ax^2 + 2hxy + by^2 = 0$ are parallel if:

(A) $h^2 - ab = 0$ (B) $h^2 - ab > 0$

(C) $h^2 - ab < 0$ (D) $h^2 = a + b$

Q2. The slope intercept form of equation of line is:

(A) $1 = x/a + y/b$ (B) $y = mx + c$

(C) $y = x/m + c$ (D) $y - y_1 = m(x - x_1)$

Q3. Two lines ℓ_1 and ℓ_2 with slope m_1 and m_2 are parallel if:

(A) $m_1 = -m_2$ (B) $m_1 = m_2$

(C) $m_1m_2 = -1$ (D) $m_1 = 1/m_2$

Q4. $x = 5$ is not solution of inequality:

(A) $x + 4 > 0$ (B) $2x + 3 < 0$

(C) $x - 4 > 0$ (D) $x + y > 4$

Q5. The parametric equations $x = a \cos \theta$, $y = a \sin \theta$ represent equation of:

(A) Circle
(B) Ellipse
(C) Hyperbola
(D) Parabola

Q6. The length of tangent from (0 , 1) to circle $x^2 + y^2 + 6x - 3y + 3 = 0$ is:

(A) 2
(B) –2
(C) 1
(D) 3

Q7. For parabola value of eccentricity e is:

(A) $e = 0$
(B) $e < 1$
(C) $e > 1$
(D) $e = 1$

Q8. $\hat{i} \cdot (\hat{j} \times \hat{j}) =$:

(A) 1
(B) i
(C) 0
(D) 2

Q9. If u is non-zero vector then $u \cdot u =$:

(A) 0
(B) 1
(C) –1
(D) u^2

Q10. A vector perpendicular to both vectors a and b is:

(A) $a \cdot b$
(B) $a \times b$
(C) $(a \cdot b)/|a|$
(D) $b \cdot a$

Q11. $\lim_{n \to \infty} (1 - 1/n)^n =$:

(A) e^{-1}
(B) e
(C) e^2
(D) $1/e^2$

Q12. Domain of $f(x) = 2 + \sqrt{(x - 1)}$, $\forall x \in R$ is:

(A) $(-1, \infty)$
(B) $(1, \infty)$
(C) $(2, \infty)$
(D) $(-2, \infty)$

Q13. If $f(x) = \cos x$ then $f'(\sin^{-1}x) =$:

(A) –sin x
(B) –x
(C) 1
(D) x

Q14. If $y = e^{2x}$ then $y_4 =$:

(A) $16\ e^{2x}$
(B) $8\ e^{2x}$
(C) $4\ e^{2x}$
(D) $-16\ e^{2x}$

Q15. $1/x \, d/dx (\sin x^2) =$:

(A) $2x \cos x^2$ (B) $2 \cos x^2$

(C) $2x \sin x^2$ (D) $\sin x^2$

Q16. If $y = 5e^{3x-4}$ then $dy/dx =$:

(A) $5 \, e^{3x}$ (B) e^{3x-4}

(C) $15 \, e^{3x-4}$ (D) $5 (3x - 4)$

Q17. $\int a/x \, dx =$:

(A) $ax + c$ (B) $a \, \ell n|x| + c$

(C) $-a/x^2 + c$ (D) $1/a \, \ell nx + c$

Q18. $\int e^x(\sin x + \cos x) \, dx =$:

(A) $e^x \cos x + c$ (B) $e^x \sin x$

(C) $e^x \sin x + c$ (D) $e^x \cos x$

Q19. $\int \sin 5x \, dx =$:

(A) $-1/a \cos x$ (B) $-1/5 \cos 5x + c$

(C) $1/5 \sin x + c$ (D) $1/5 \cos 5x + c$

Q20. Solution of different equation $dy/dx = -y$ is:

(A) $y = ce^x$ (B) $y = ce^{-x}$

(C) e^x (D) $1/c \, e^{-x}$

Mathematics (Subjective) Group - I

Time Allowed: 02:30 Hours | **Maximum Marks:** 80 | **Paper Code:** 8197

Note: SECTION - I

Q2. (Marks 16) Attempt any EIGHT parts:

(i) Find the domain and range of $g(x) = \sqrt{(x^2 - 4)}$

(ii) Find $f^{-1}(x)$ if $f(x) = (2x + 1)/(x - 1)$

(iii) Find Lim $x \to a \, (x^n - a^n)/(x^m - a^m)$

(iv) Find Lim x→0 (e^(1/x) – 1)/(e^(1/x) + 1), x > 0

(v) If $y = x^4 + 2x^2 + 2$, prove that $dy/dx = 4x \sqrt{(y - 1)}$

(vi) Differentiate sin x w.r.t. cot x

(vii) Find dy/dx if $x^2 - 4xy - 5y = 0$

(viii) If $f(x) = \ell n \sqrt{(e^{2x} + e^{-2x})}$, find f'(x)

(ix) If $y = \ell n(\tanh x)$, find dy/dx

(x) If $y = x^2 \ell n(1/x)$, find dy/dx

(xi) If $x = a(\cos t + \sin t)$, $y = a(\sin t - t \cos t)$, find dy/dx

(xii) Apply Maclaurin series prove that $e^{2x} = 1 + 2x + 4x^2/2! + \ldots\ldots$

Q3. (Marks 16) Attempt any EIGHT parts:

(i) Use differential to approximate the value of $\sqrt{17}$

(ii) Evaluate $\int x \sqrt{(x^2 - 1)}\, dx$

(iii) Evaluate $\int(\sqrt{x} + 1/\sqrt{x})\, dx$

(iv) Evaluate $\int \tan^2 x\, dx$

(v) Evaluate $\int \sec^2 x/\sqrt{(\tan x)}\, dx$

(vi) Evaluate $\int \ell n x\, dx$

(vii) Evaluate $\int_0\hat{}(\pi/6)\, x \cos x\, dx$

(viii) Solve the differential equation $y\, dx + x\, dy = 0$

(ix) Find the coordinates of the point that divides the join of A(–6, 3) and B(5, –2) in the ratio 2 : 3 internally.

(x) By means of slopes [illegible in source scan] the points (4, –5), (7, 5) and (10, 15) lie on the same line.

(xi) Find [illegible in source scan] with y-intercept: –7 and slope: –5.

(xii) [illegible in source scan] $y\, 2x^2 + 3xy - 5y^2 = 0$

Note: (Continued P/2)

Q4. (Marks 18) Attempt any NINE parts:

(i) Graph the solution set of $3x - 2y \geq 6$

(ii) Find equation of circle with center at $(\sqrt{2}, -3\sqrt{3})$ and radius $2\sqrt{2}$

(iii) Find length of tangent from point P (–5, 10) to circle $5x^2 + 5y^2 + 14x + 12y - 10 = 0$

(iv) Find vertex and directrix of parabola $x^2 = -16y$

(v) Find equation of parabola with focus (–3, 1) and directrix $x = 3$

(vi) Find center and foci of $x^2/4 - y^2/9 = 1$

(vii) Find eccentricity and vertex of $y^2/16 - x^2/9 = 1$

(viii) Write the vector PQ in the form $x\, i + y\, j$, P(2, 3), Q(6, –2)

(ix) Find a unit vector in the direction of $v = 2i - j$

(x) Find a vector whose magnitude is 4 and is parallel to $2i - 3j + 6k$

(xi) Find a real number α so that $u = \alpha i + 2\alpha j - k$, $v = i + \alpha j + 3k$ are perpendicular.

(xii) Compute $b \times a$ if $a = 2i + j - k$, $b = i - j + k$

(xiii) Prove that $u \cdot (v \times w) + v \cdot (w \times u) + w \cdot (u \times v) = 3u \cdot (v \times w)$

> Note: SECTION - II Attempt any THREE questions. Each question carries 10 marks.

Q5(a). (Marks 05) Evaluate $\lim_{\theta \to 0} (\tan\theta - \sin\theta)/\sin^3\theta$

Q5(b). (Marks 05) Show that $dy/dx = y/x$ if $y/x = \tan^{-1} x/y$

Q6(a). (Marks 05) Solve $\int e^{-x} \sin 2x \, dx$

Q6(b). (Marks 05) Find the angles of the triangle whose vertices are A(–5, 4), B(–2, –1) and C(7, –5)

Q7(a). (Marks 05) Evaluate $\int_0^{(\pi/4)} \cos^4 t \, dt$

Q7(b). (Marks 05) Maximize $f(x, y) = x + 3y$ subject to the constraints: $2x + 5y \leq 30$; $5x + 4y \leq 20$; $x \geq 0, y \geq 0$

Q8(a). (Marks 05) Write an equation of circle passing through the points A(–7, 7), B(5, –1), C(10, 0)

Q8(b). (Marks 05) Given force $\vec{F} = 2i + j - 3k$ acting at a point A(1, –2, 1) find the moment of $\vec{F}$ about the point B(2, 0, –2)

Q9(a). (Marks 05) Show that $y = \ell nx/x$ has maximum value at $x = e$

Q9(b). (Marks 05) Show that the ordinate at any point P of the parabola is a mean proportional between the length of the latus rectum and the abscissa of P.

GROUP – II

Mathematics (Objective) Group - II

Time Allowed: 30 Minutes	**Maximum Marks:** 20	**Paper Code:** 8198

> Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. The altitudes of a triangle are:

(A) Concurrent
(B) Parallel
(C) Perpendicular
(D) Imaginary

Q2. The line y = 3x passes through:

(A) Origin
(B) (4, 3)
(C) (3, 1)
(D) (0, 3)

Q3. A quadrilateral having two parallel and two non-parallel sides is called:

(A) Parallelogram
(B) Rhombus
(C) Trapezium
(D) Triangle

Q4. The maximum or minimum values of objective function occur at corner points of the feasible region, is called:

(A) The theorem of linear programming
(B) Feasible theorem
(C) Optional theorem
(D) Convex

Q5. The focus of $y^2 = -4ax$ is:

(A) (0, 0)
(B) (a, 0)
(C) (–a, 0)
(D) (a, a)

Q6. The eccentricity of parabola is:

(A) 0
(B) 1
(C) Less than one
(D) Not defined

Q7. The center of $x^2/a^2 - y^2/b^2 = 1$ is:

(A) (0, 0)
(B) (a, 0)
(C) (0, b)
(D) (a, b)

Q8. If u, v and w are coplaner vectors than the volume of the parallelopiped so formed is:

(A) 1
(B) 0
(C) $|u \times v \times w|$
(D) $\sqrt{(u^2 + v^2 + w^2)}$

Q9. The magnitude of u = i + j is:

(A) $2\sqrt{(i^2 + j^2)}$
(B) 2
(C) $\sqrt{2}$
(D) $(\hat{i} + \hat{j})/\sqrt{2}$

Q10. The unit vector in the direction of $2\hat{i} - \hat{j}$ is:

(A) 5
(B) $\sqrt{5}$
(C) (2i – j)/5
(D) (2i – j)/$\sqrt{5}$

Q11. A function C : R → R defined by C(x) = 2 for all x ∈ R is called:

(A) Domain
(B) Range
(C) Constant function
(D) Objective function

Q12. For the function f(x) = x^n, lim x→a $(x^n - a^n)/(x - a)$ = :

(A) nx^{n-1}
(B) na^{n-1}
(C) 0
(D) ∞

Q13. If f(x) = |x| then f'(0) = :

(A) 0
(B) 1
(C) –1
(D) Does not exist

Q14. The sum of two integers is 9. If one integer is x, then other will be:

(A) 9x
(B) 9 – x
(C) x – 9
(D) x + 9

Q15. The derivative of arc cos x is:

(A) $-1/\sqrt{(1 - x^2)}$
(B) $1/\sqrt{(1 + x^2)}$
(C) $1/\sqrt{(x^2 - 1)}$
(D) $-1/\sqrt{(x^2 - 1)}$

Q16. The first term of Taylor Series Expansion of ℓn(1 + x) at x = 2 is:

(A) ℓn 3
(B) ℓn 2
(C) ℓn 1
(D) ℓn 0

Q17. ∫tan x dx = :

(A) $\sec^2 x$
(B) sec x tan x
(C) ℓn sec x
(D) ℓn cos x

Q18. The suitable substitution to integrate $\sqrt{(x^2 - a^2)}$:

(A) x = a sin θ
(B) x = a cos θ
(C) x = a sec θ
(D) x = a tan θ

Q19. $\int e^{ax}[af(x) + f'(x)]\, dx$ = :

(A) $e^{ax}f(x)$
(B) $e^{ax} \cdot a \cdot f'(x)$
(C) $e^{ax}f'(x)$
(D) a f'(x)

Q20. $\int 2/(x + 2)\, dx$ = :

(A) ℓn|x + 2|
(B) $\ell n|x + 2|^2$
(C) $1/(\ell n|x + 2|)$
(D) 2

Mathematics (Subjective) Group - II

Time Allowed: 02:30 Hours | **Maximum Marks:** 80 | **Paper Code:** 8198

Note: SECTION - I

Q2. (Marks 16) Attempt any EIGHT parts:

(i) Determine whether the function is even or odd for $f(x) = x^2/x^3 + 6$

(ii) Without finding inverse state domain and range of $f(x) = (x - 1)/(x - 4)$, $x \neq 4$

(iii) Evaluate $\lim_{\theta \to 0} (1 - \cos \theta)/\theta$

(iv) Show that $x = at^2$, $y = 2at$ represent parametric equation of $y^2 = 4ax$

(v) Differentiate w.r.t. 'x' if $y = (x^2 + 1)/(x^2 - 3)$

(vi) Differentiate $x^2 - 1/x^2$ w.r.t. x^4

(vii) Find dy/dx if y = xe^(sin x)

(viii) Find y_2 if $x^3 - y^3 = a^3$

(ix) Prove that $d/dx (\cot^{-1}x) = -1/(1 + x^2)$

(x) Differentiate w.r.t. x, $\sin^{-1}\sqrt{(1 - x^2)}$

(xi) Find dy/dx if y = e^(–2x) sin 2x

(xii) Find y_4 if $y = \ell n(x^2 - 9)$

Q3. (Marks 16) Attempt any EIGHT parts:

(i) Using differentials find dy/dx when $y/x - \ell nx = \ell nc$

(ii) Evaluate $\int (3 - \cos 2x)/(1 + \cos 2x)\, dx$

(iii) Evaluate the indefinite integral $\int (\sqrt{\theta} - 1)^2/\sqrt{\theta}\, d\theta$

(iv) Evaluate $\int \sqrt{2}/(\sin x + \cos x)\, dx$

(v) Find the antiderivative of $x\ell nx$

(vi) Evaluate the given integral $\int \sec^4 x\, dx$

(vii) Solve the differential equation sec x + tan y dy/dx = 0

(viii) Find the area, above the x-axis and under the curve $y = 5 - x^2$ from x = –1 to x = 2

(ix) Show that the points A(0, 2), B(√3, –1) and C(0, –2) are vertices of a right triangle.

(x) Convert the given equation into normal form: 15y – 8x + 3 = 0

(xi) Find the interior angles (any two) of the triangle whose vertices are A(2, –5), B(–4, –3), C(–1, 5)

(xii) Find an equation of each of the lines represented by $2x^2 + 3xy - 5y^2 = 0$

> Note: (Continued P/2)

Q4. (Marks 18) Attempt any NINE parts:

(i) Graph the solution set of $2x + y \leq 6$ by shading.

(ii) Find the value of α so that the vectors $\alpha i + j$, $i + j + 3k$ and $2i + j - 2k$ are coplaner.

(iii) If 0 is the origin and $\vec{OP} = \vec{AB}$. Find the point P when A and B are (–3, 7) and (1, 0) respectively.

(iv) If $v = 3i - 2j + 2k$ and $w = 5i - j + 3k$, then find $|3v + w|$

(v) Find a unit vector perpendicular to plane containing a and b and $a = -i - j - k$, $b = 2i - 3j + 4k$

(vi) Compute cross product, $a \times b$, $b \times a$, if $a = -4i + j - 2k$ and $b = 2i + j + k$

(vii) Find work done, if the point at which the constant force $F = 4i + 3j + 5k$ is applied to an object moves from $P_1(3, 1, -2)$ to $P_2(2, 4, 6)$

(viii) Find an equation of circle with center at $(\sqrt{2}, -3\sqrt{3})$ and radius is $2\sqrt{2}$

(ix) Find length of tangent drawn from point P(–5, 4) to the circle $5x^2 + 5y^2 - 10x + 15y - 131 = 0$

(x) Find focus and vertex of the parabola $x^2 = 4(y - 1)$

(xi) Find center and eccentricity of $4y^2 + 12y - x^2 + 4x + 1 = 0$.

(xii) Define circle and just write its standard equation.

(xiii) Find equation of tangent to the circle $4x^2 + 3y^2 + 5x - 13y + 2 = 0$ at (1, 10/3).

> Note: SECTION - II Attempt any THREE questions. Each question carries 10 marks.

Q5(a). (Marks 05) If $f(x) = \{(\sqrt{(2x + 5)} - \sqrt{(x + 7)})/(x - 2), x \neq 2; k, x = 2\}$, find the value of k so that f is continuous at x = 2

Q5(b). (Marks 05) Differentiate ab-initio w.r.t. 'x', $\sin \sqrt{x}$

Q6(a). (Marks 05) Evaluate $\int \sqrt{2}/(\sin x + \cos x)\, dx$

Q6(b). (Marks 05) Find the condition that lines $y = m_1x + c_1$, $y = m_2x + c_2$ and $y = m_3x + c_3$ are concurrent.

Q7(a). (Marks 05) Evaluate $\int_0^{(\pi/4)} (\sin x - 1)/\cos^2 x\, dx$

Q7(b). (Marks 05) Maximize $f(x, y) = 2x + 5y$ subject to the constraints: $2y - x \leq 8$, $x - y \leq 4$, $x \geq 0$, $y \geq 0$

Q8(a). (Marks 05) Find an equation of a circle which passes through A(–3, 1) with radius 2 and center at 2x – 3y + 3 = 0

Q8(b). (Marks 05) If a + b + c = 0 then prove that a × b = b × c = c × a

Q9(a). (Marks 05) If $y = a\cos(\ell nx) + b\sin(\ell nx)$ prove that $x^2 d^2y/dx^2 + x\, dy/dx + y = 0$

Q9(b). (Marks 05) Find the center, foci and vertices of hyperbola $9x^2 – y^2 – 36x – 6y + 18 = 0$